

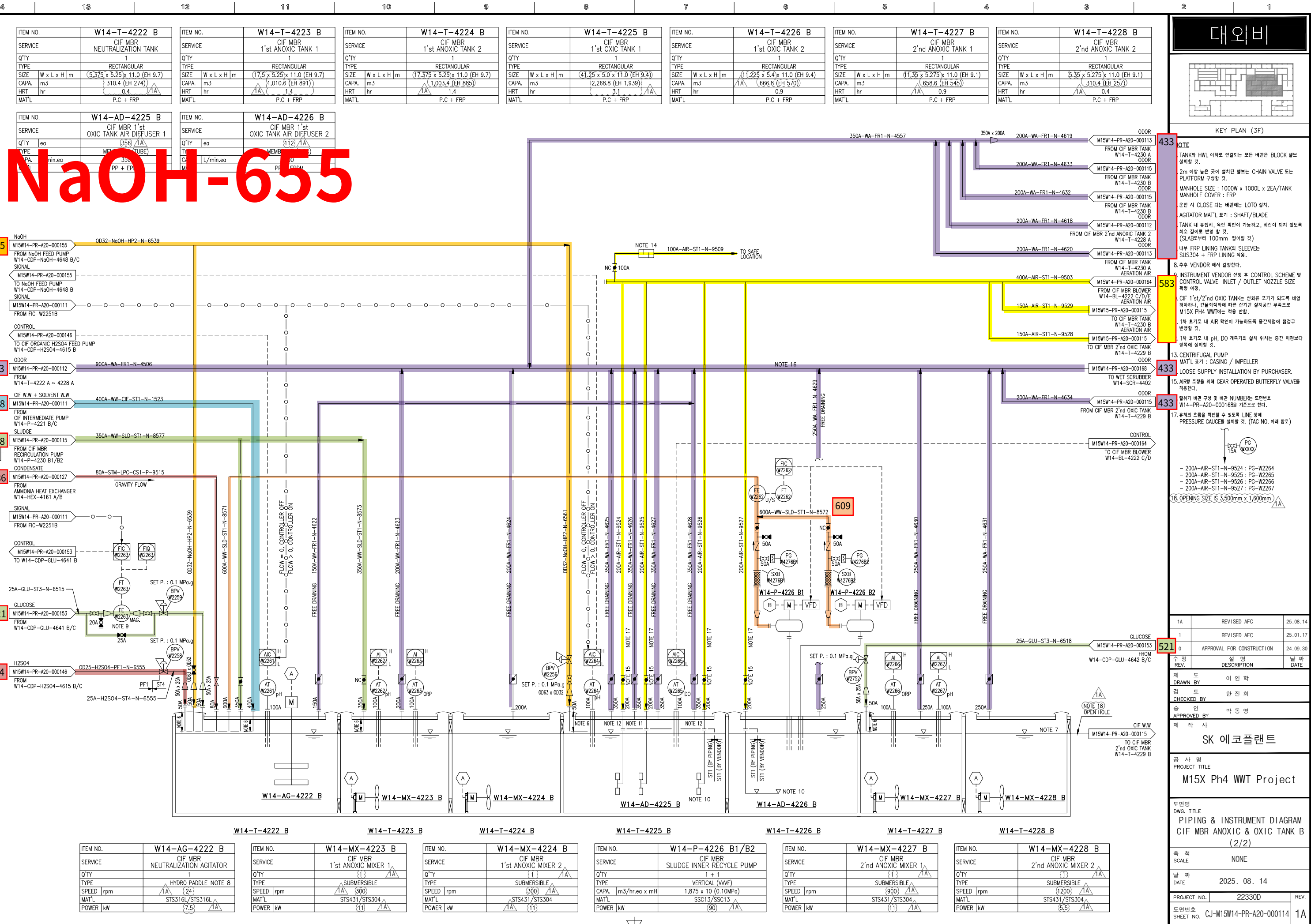
DRAWING LIST FOR TEST PACKAGE

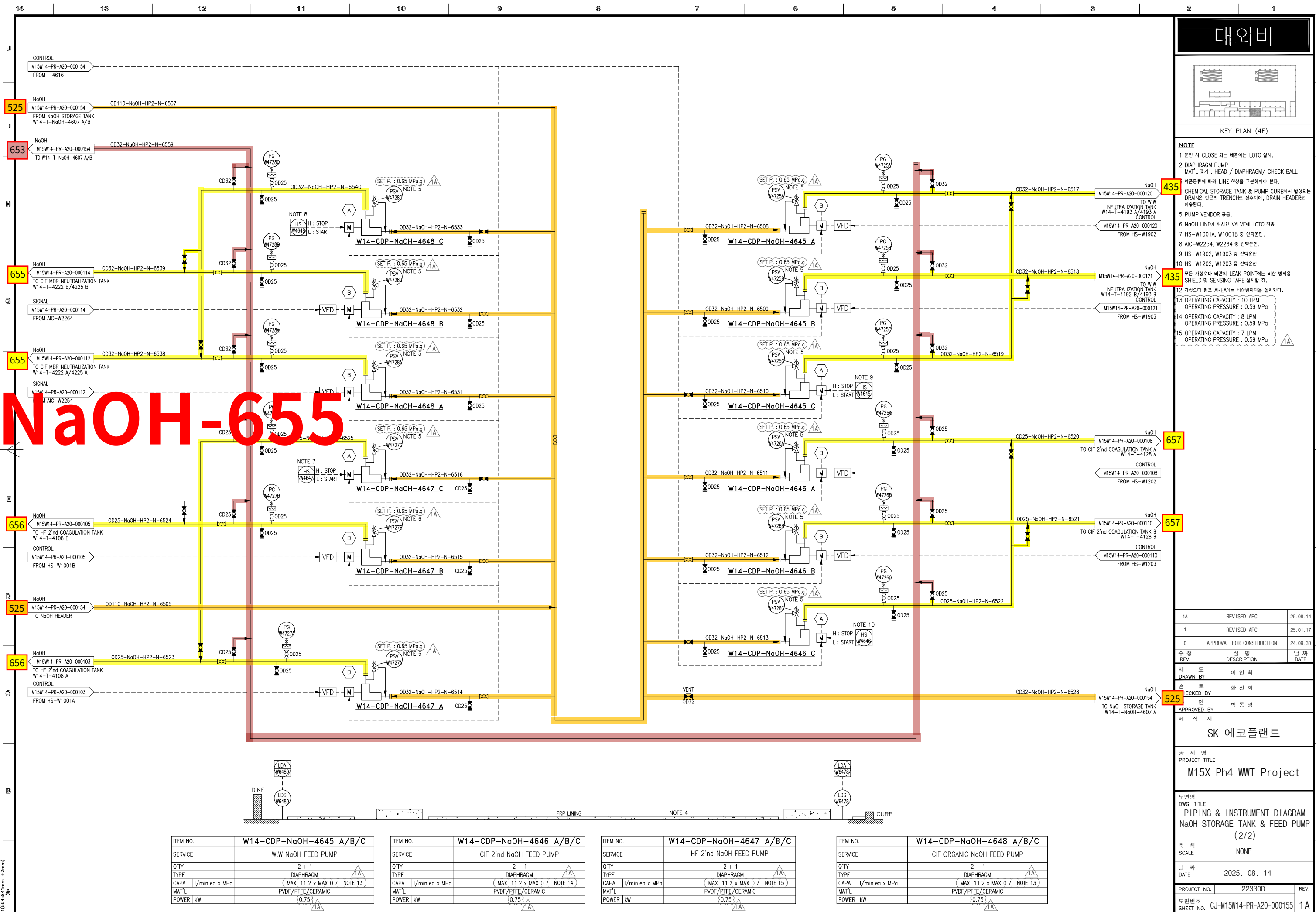
NAOH-655

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	655	4FA	OD32-NaOH-HP2-N-6538-01	1 OF 2	0	CJ-M15W14-PR-A20-000155		
2	655	4FA	OD32-NaOH-HP2-N-6538-02	2 OF 2	0	CJ-M15W14-PR-A20-000155, 112		
3	655	4FA	OD32-NaOH-HP2-N-6539-01	1 OF 2	0	CJ-M15W14-PR-A20-000155		
4	655	4FA	OD32-NaOH-HP2-N-6539-02	2 OF 2	0	CJ-M15W14-PR-A20-000155, 114		
5	655	4FA	OD32-NaOH-HP2-N-6540-01	1 OF 1	0	CJ-M15W14-PR-A20-000155		
6	655	4FA	OD32-NaOH-HP2-N-6560-01	1 OF 1	0	CJ-M15W14-PR-A20-000112		
7	655	4FA	OD32-NaOH-HP2-N-6561-01	1 OF 1	0	CJ-M15W14-PR-A20-000114		
8	655	4FA	OD32-NaOH-HP2-N-6559-02	2 OF 2	0	CJ-M15W14-PR-A20-000155		
9								
10								
11								
12								





대외비

KEY PLAN (4F)

NOTE

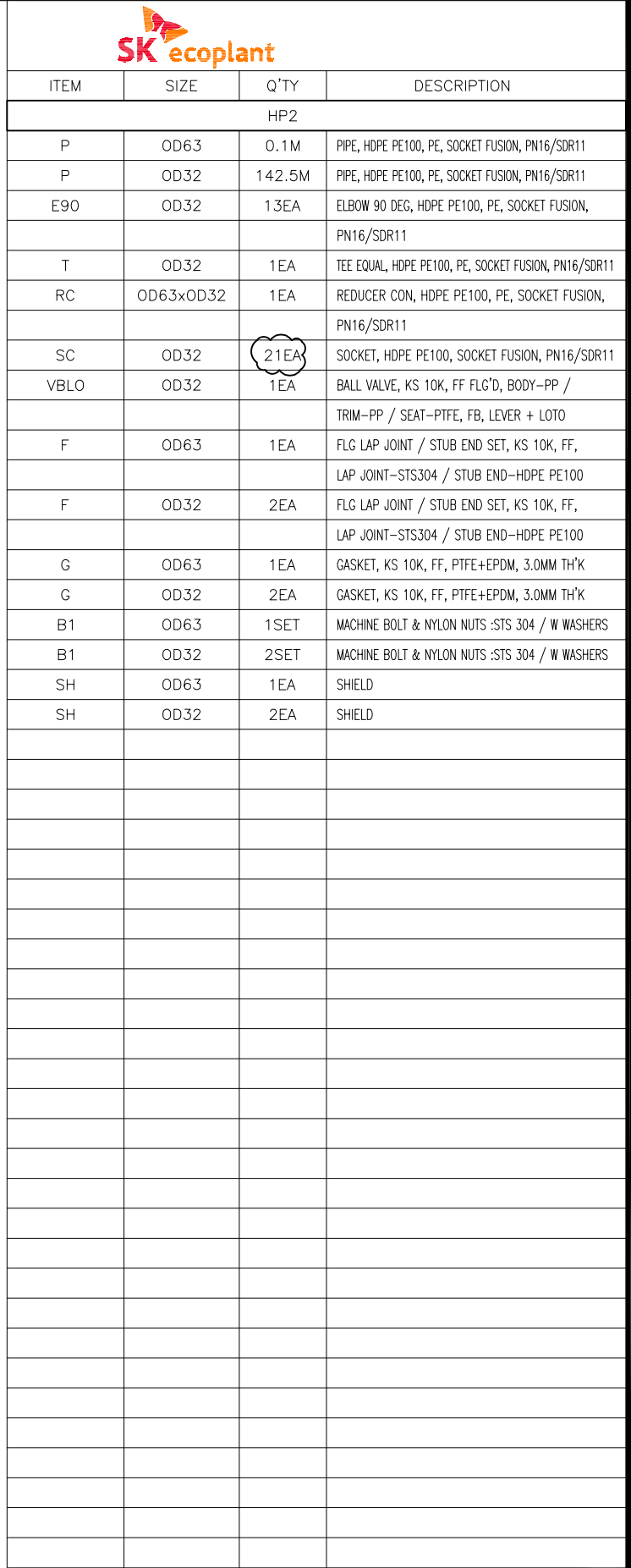
1. 운전 시 CLOSE 되는 배관에는 LOTO 실시.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. NaOH LINE에 위치한 VALVE에 LOTO 적용.
7. HS-W1001A, W1001B 중 선택운전.
8. AIC-W2254, W2264 중 선택운전.
9. HS-W1902, W1903 중 선택운전.
10. HS-W1202, W1203 중 선택운전.

모든 가성소터 배관의 LEAK POINT에는 비산 방지를 SHIELD 및 SENSING TAPE 설치할 것.

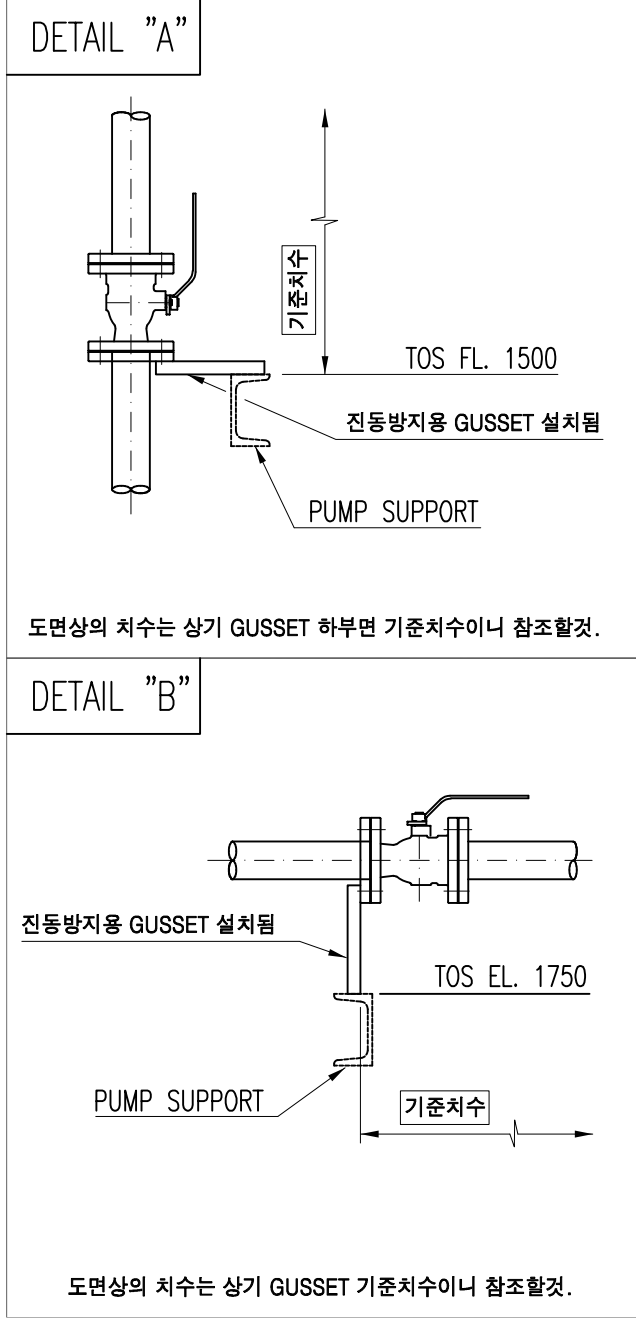
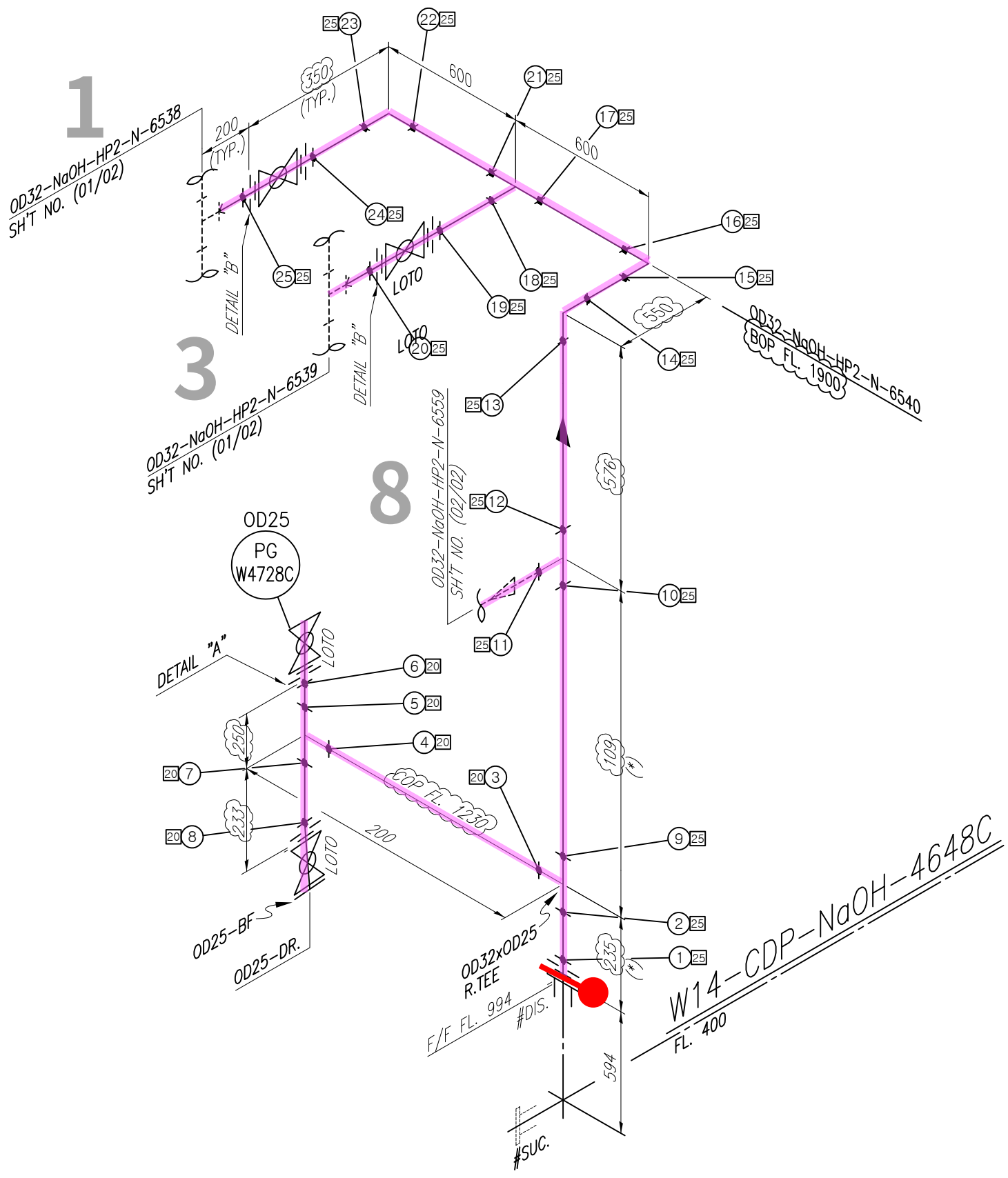
12. 가성소터 배관 AREA에는 비산방지막을 설치한다.
13. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.59 MPa
14. OPERATING CAPACITY : 8 LPM
OPERATING PRESSURE : 0.59 MPa
15. OPERATING CAPACITY : 7 LPM
OPERATING PRESSURE : 0.59 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인 학
DRAWN BY		
검	토	한 진 희
CHECKED BY		
APPROVED BY	박 동 영	
제	작	사
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
NaOH STORAGE TANK & FEED PUMP		
(2/2)		
속		
척		
SCALE		
NONE		
날		
짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		
SHEET NO.		CJ-M15W14-PR-A20-000155
		1A

ITEM NO.	W14-CDP-NaOH-4645 A/B/C	ITEM NO.	W14-CDP-NaOH-4646 A/B/C	ITEM NO.	W14-CDP-NaOH-4647 A/B/C	ITEM NO.	W14-CDP-NaOH-4648 A/B/C
SERVICE	W.W NaOH FEED PUMP	SERVICE	CIF 2'nd NaOH FEED PUMP	SERVICE	HF 2'nd NaOH FEED PUMP	SERVICE	CIF ORGANIC NaOH FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa
MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC
POWER	kw	POWER	kw	POWER	kw	POWER	kw



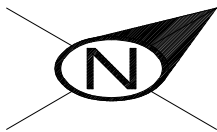
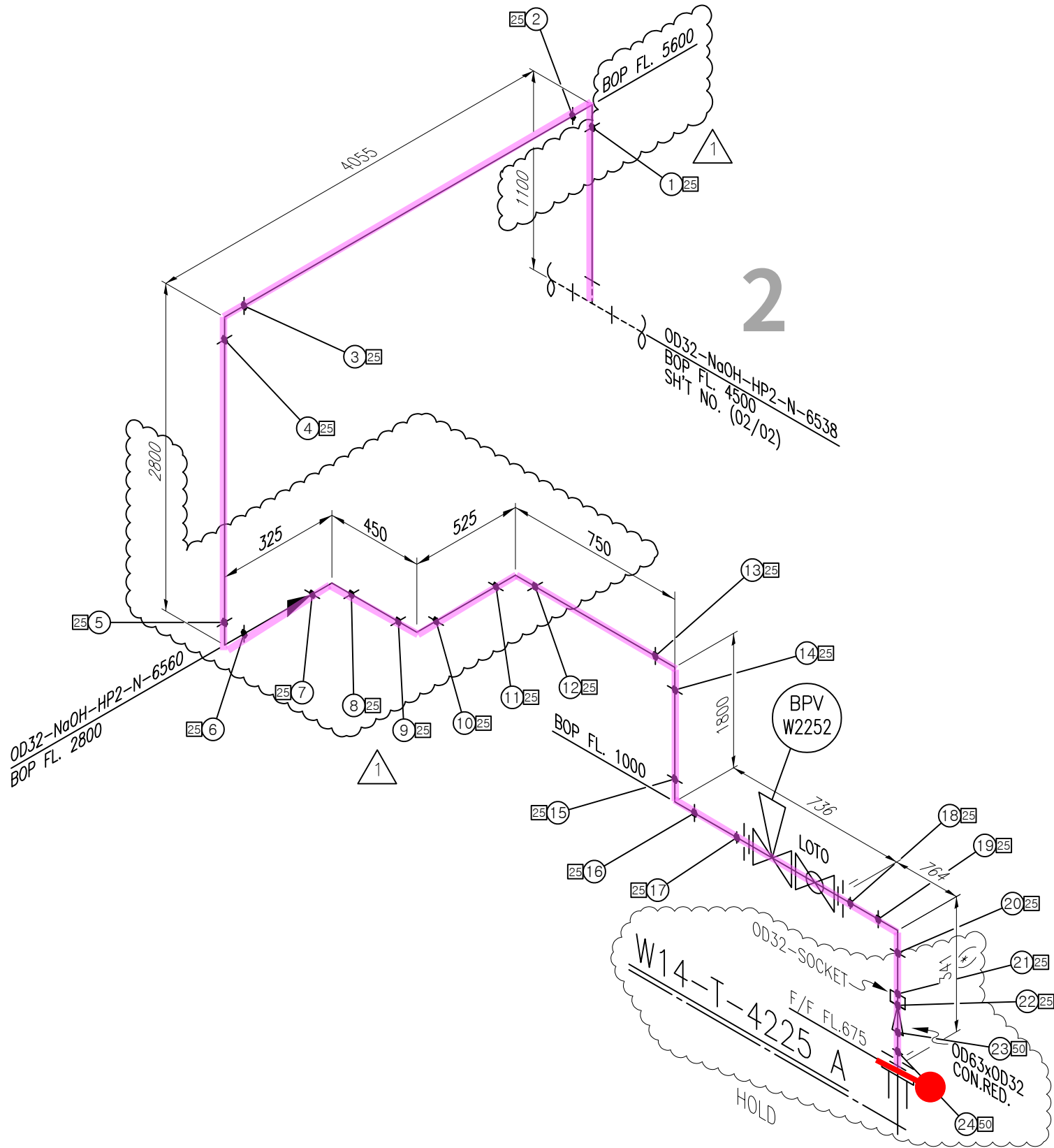
CLIENT		 2	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
NaOH		22330D	
LINE NO.	SH'T NO.	REV.	
0D32-NaOH-HP2-N-6538	02 / 02	1	



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR		PROJECT NAME	
	THICK.	—	DESIGN	6.8	60	UNIT	4FA									M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ—M15W14—PI—A30—040012—A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	NaOH	JOB NO.	
						REF. P & ID CJ—M15W14—PR—A20—000155											

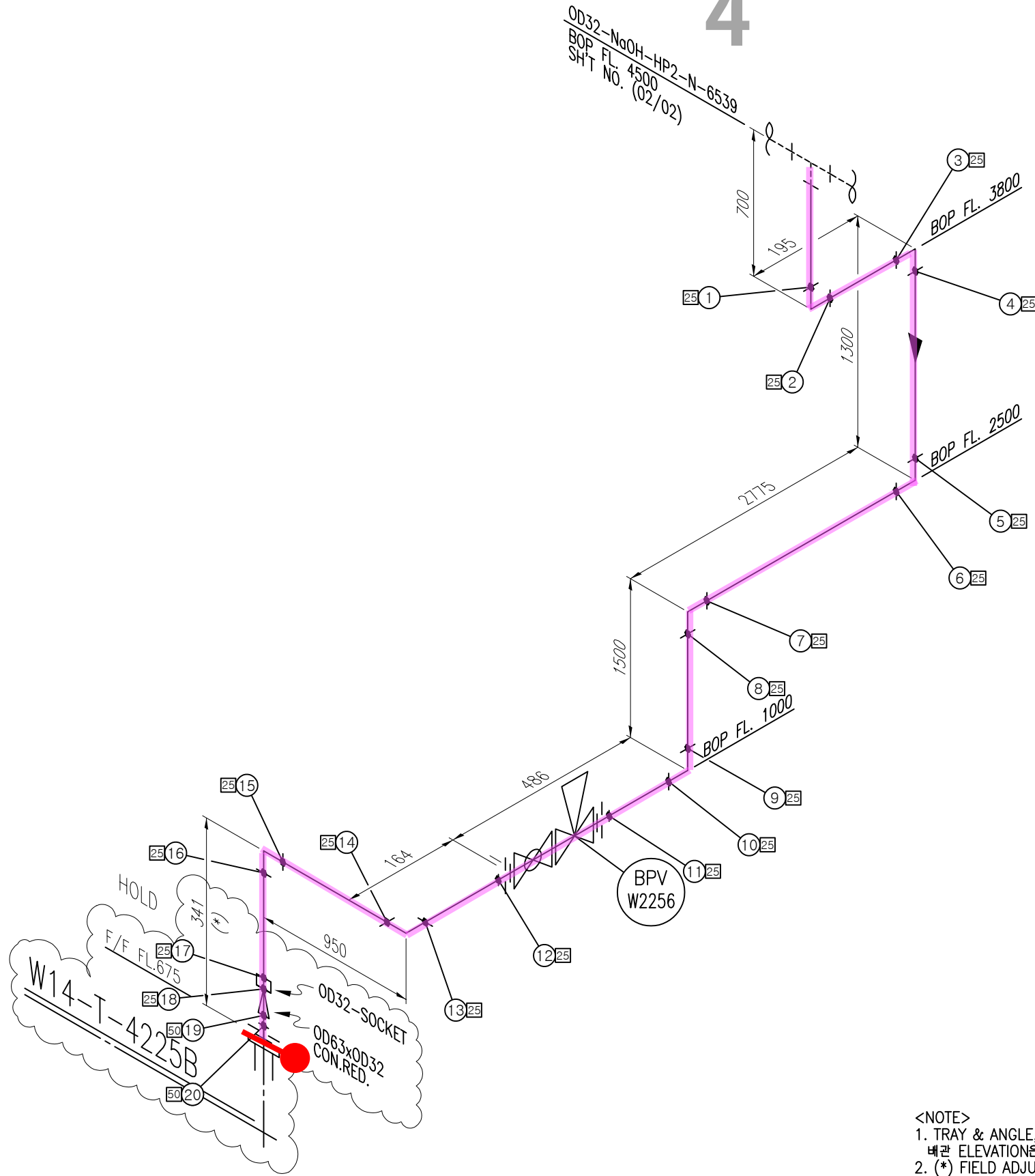
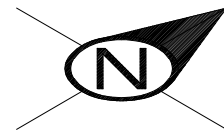


ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD63	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD32	14.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	9EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION,
RC	OD63xOD32	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
SC	OD32	1EA	SOCKET, HDPE PE100, SOCKET FUSION, PN16/SDR11
VBLO	OD32	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
F	OD63	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
F	OD32	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
G	OD63	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B1	OD63	1SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD32	3SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	OD63	1EA	SHIELD
SH	OD32	3EA	SHIELD
BPV	OD32	1EA	BACK PRESSURE VALVE

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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.8	60	UNIT	4FA									M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-040002-C			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D		
						REF. P & ID CJ-M15W14-PR-A20-000112		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD32-NaOH-HP2-N-6560	SH'T NO. 01 / 01	REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	6.8	60	UNIT	4FA							SK ecoplant	M15X Ph4 Project
			OPERATING	6.1	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040005-C							COMMODITY	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000114	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.
														OD32-NaOH-HP2-N-6561	01 / 01
															REV. 0

